



CFD-THM coupling in Hippo

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- Motivation
- Intro to 0D/1D thermal hydraulic analysis
- MOOSE's thermal hydraulics module
- Coupling with CFD using Hippo
 - Overlapping domain vs Non-overlapping domain

Motivation

- Improving fidelity of components known to be difficult to capture using system code
- Capturing system behaviour while also capturing detailed response of system components
 - Reactor transient: e.g. pump or turbine failures
- Below is an example for a test facility for an LMFR

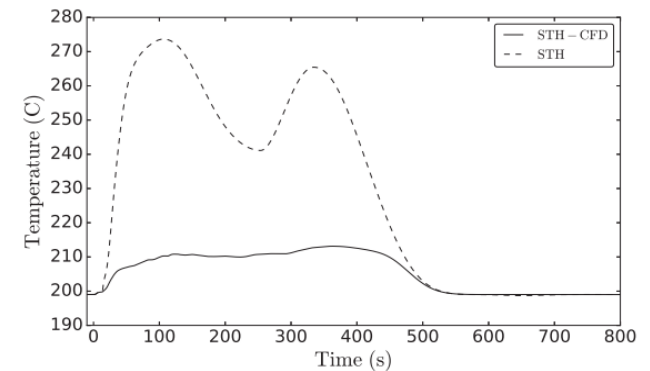
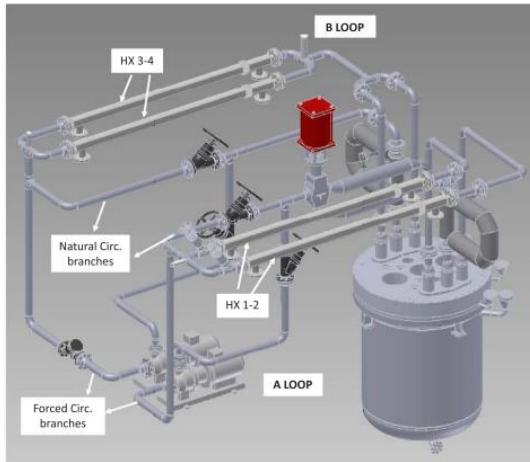
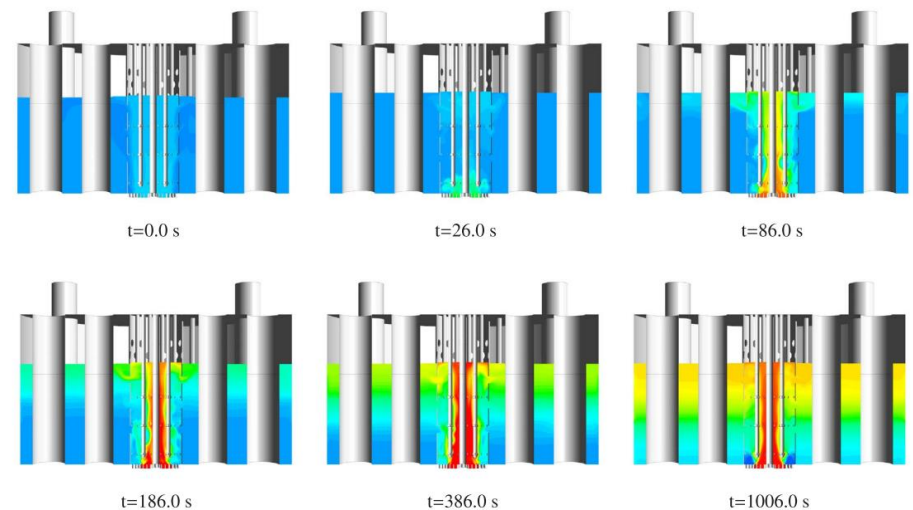


Fig. 17. LOF transient: core bypass region outlet temperature.



Overview of thermal hydraulic analysis

- Key equations (simplified – constant properties and area)

- Momentum

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} \right) = -\frac{\partial p}{\partial x} + \rho g_x - \frac{f_D}{2D_h} \rho u |u|$$

- Energy

$$\rho c_p \left(\frac{\partial T_f}{\partial t} + u \frac{\partial T_f}{\partial x} \right) = \frac{h_w P_h}{A} (T_w - T_f) + \dot{q}'''$$

- Closures need to be found for:

- Friction factor f_D : correlations for the Darcy friction factor
- Heat transfer coefficient, h_w : Nusselt number correlations

MOOSE's thermal hydraulics module

- Built on top of the MOOSE framework. Introduces
 - Components:
 - FlowChannel1Phase
 - Outlet1Phase
 - InletMassFlowRateTemperature1Phase
 - ControlLogic
 - SetComponentRealValue: dynamically sets the parameters of components
 - Closures
 - Select Nusselt number and friction factor correlations

```
[ControlLogic]
[mdot_inlet]
  type = SetComponentRealValueControl
  component = inlet
  parameter = m_dot
  value = mdot_inlt
[]

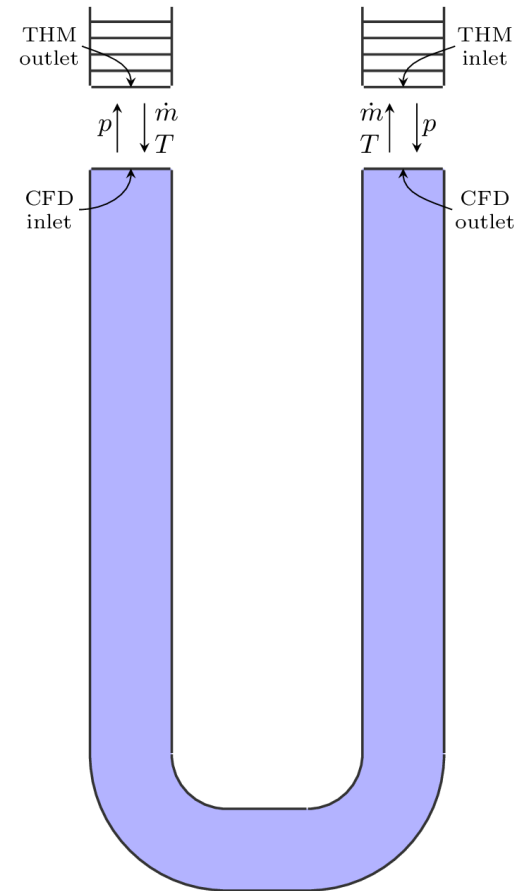
[Postprocessors]
[mdot_inlt]
  type = ParsedPostprocessor
  expression = 'if(t<0.6, 3.92, if(t<0.8, 3.92+19.6*(t-0.6), 7.84))'
  use_t = true
  execute_on = "INITIAL TIMESTEP_BEGIN"
[]

[Components]
[inlet]
  type = InletMassFlowRateTemperature1Phase
  T = 300
  m_dot = 3.92
  input = pipe1:in
[]
[pipe1]
  type = FlowChannel1Phase
  position = '0 0 -1'
  orientation = '0 0 1'
  length = 3
  n_elems = 100
  A = ${A}
  D_h = ${D}
[]
[outlet]
  type = Outlet1Phase
  input = pipe1:out
  p = ${press}
[]

[Closures] # defines friction factors and heat transfer coefficients
[thm_closures]
  type = Closures1PhaseTHM # default Churchill friction factor, DB HTC
[]
```

Coupling thermal hydraulics with CFD

- There are two ways of coupling TH with CFD
 - Overlapping domain
 - Generally superior particularly for string transients
 - I hope to implement this in the future
 - Non-overlapping domain
 - A little simpler to implement but tends to be less stable
 - This can be done just using Hippo components we've already introduced
- From OpenFOAM to MOOSE
 - Foam-based Postprocessors
- From MOOSE to OpenFOAM
 - Postprocessor-based BCs
 - Mapped inlets



Example: Flow through a pipe

- We will do a very simple example of CFD/THM coupling
 - We will compare THM-only with a coupled simulation
- We will consider a 3 meter pipe in two scenarios
 - Thermal transient
 - Mass flow rate transient
- For the coupled simulation
 - 0-1m: THM
 - 1-2m: CFD
 - 2-3m: THM